

TOTAL OUTREACHES

11,162

Complete call logs analyzed

SUBSCRIBED DEPOSITS

5,289

47.38% average conversion rate

TOP CONVERSION MONTH

Dec

90.91% subscription rate

STRONGEST DRIVER

Duration

36.87% predictive importance

1. Introduction

In retail banking, term deposits represent a stable and low-cost source of funding, which banks can deploy for lending activities to drive profitability. However, retail banks face intense competition and rising customer acquisition costs (CAC). Direct marketing campaigns, primarily conducted via telephone calls, remain a primary channel for selling term deposits. While direct marketing is personal, it is often inefficient, resulting in low conversion rates, customer fatigue, and high operational costs.

This report analyzes historical campaign data from a retail bank to understand customer behavior and optimize the conversion rate of term deposit subscriptions. By leveraging exploratory data analysis (EDA) and machine learning (Random Forest feature importance), we identify key drivers of subscription success and provide data-driven recommendations to maximize campaign return on investment (ROI).

2. Business Problem Statement

Direct telephone marketing campaigns suffer from low efficiency. In our dataset, comprising 11,162 customer contacts, the average subscription conversion rate is 47.38%. This means 52.62% of calls result in rejection. The bank's outreach is currently unsegmented and untargeted, placing high volumes of cold calls (especially in certain months like May) that create campaign fatigue, increase call center overheads, and damage brand goodwill.

The business challenges we solve in this report are:

- **Subscription Determinants:** Why do some customers subscribe to term deposits while others do not?
- **Target Segments:** Which customer groups are more likely to respond positively to marketing campaigns?
- **Operational Inefficiencies:** How can we reduce call center volume while keeping the total number of deposits high?

3. Dataset Description & Methodology

The dataset analyzed is `bank.csv`, which contains 11,162 rows and 17 columns, with no missing values. The target variable is `deposit` (binary: yes/no), representing whether the client subscribed to a term deposit. Features encompass client demographic details, financial profiles, campaign contact details, and historical campaign records.

Our analytical workflow is as follows:

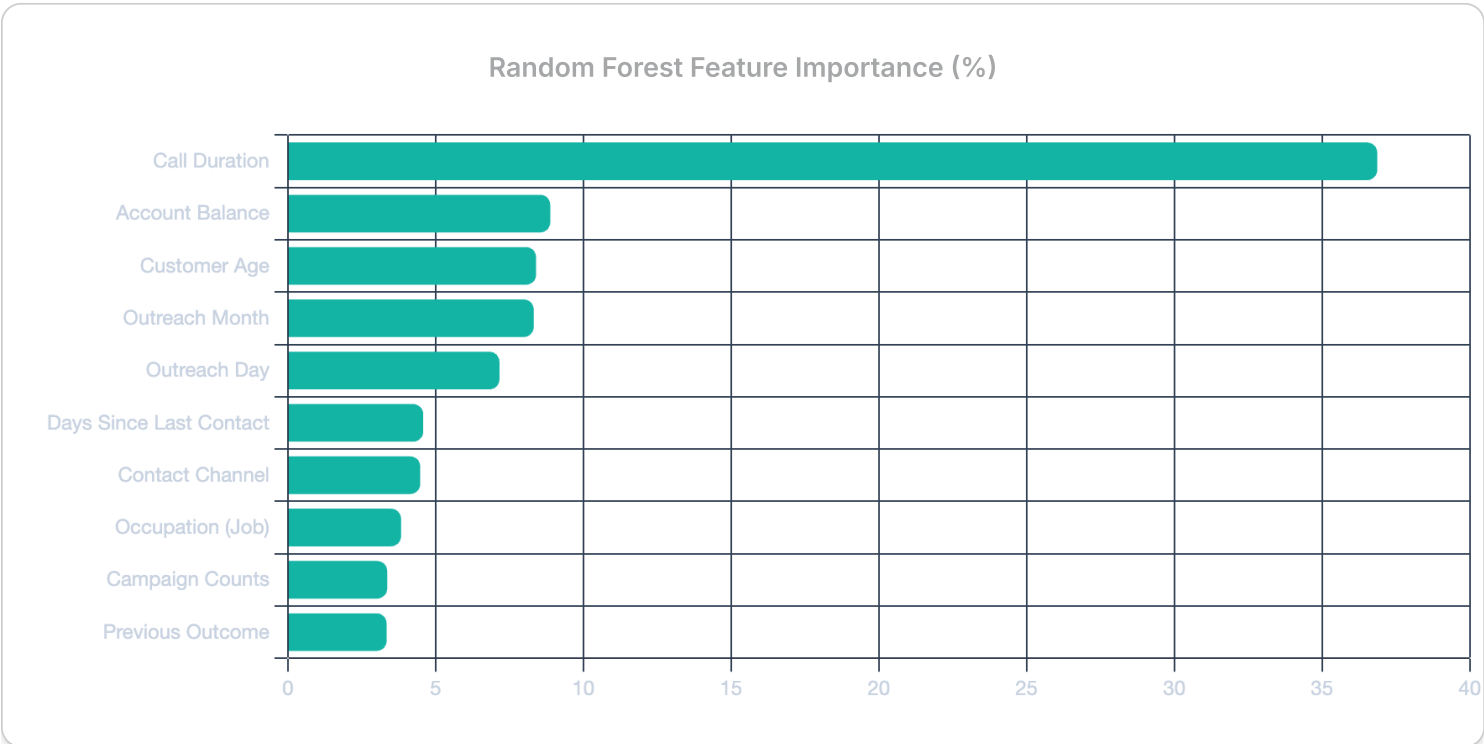
- Data Aggregation:** Grouping categorical features to analyze conditional subscription probabilities.
- Feature Binning:** Segmenting continuous numerical variables (such as age, account balance, call duration, and contact count) into quantiles/intervals to capture non-linear behaviors.
- Statistical Modeling:** Training a Random Forest Classifier to assign relative feature importance scores, isolating the top drivers of subscription success.

4. Key Findings & Data Analysis

Our exploratory analysis and predictive modeling revealed that subscription conversion is highly correlated with specific behavioral, temporal, and demographic features.

Statistical Key Drivers (Feature Importance)

Training a Random Forest Classifier on label-encoded features determined the relative importance of each feature in predicting subscriptions:

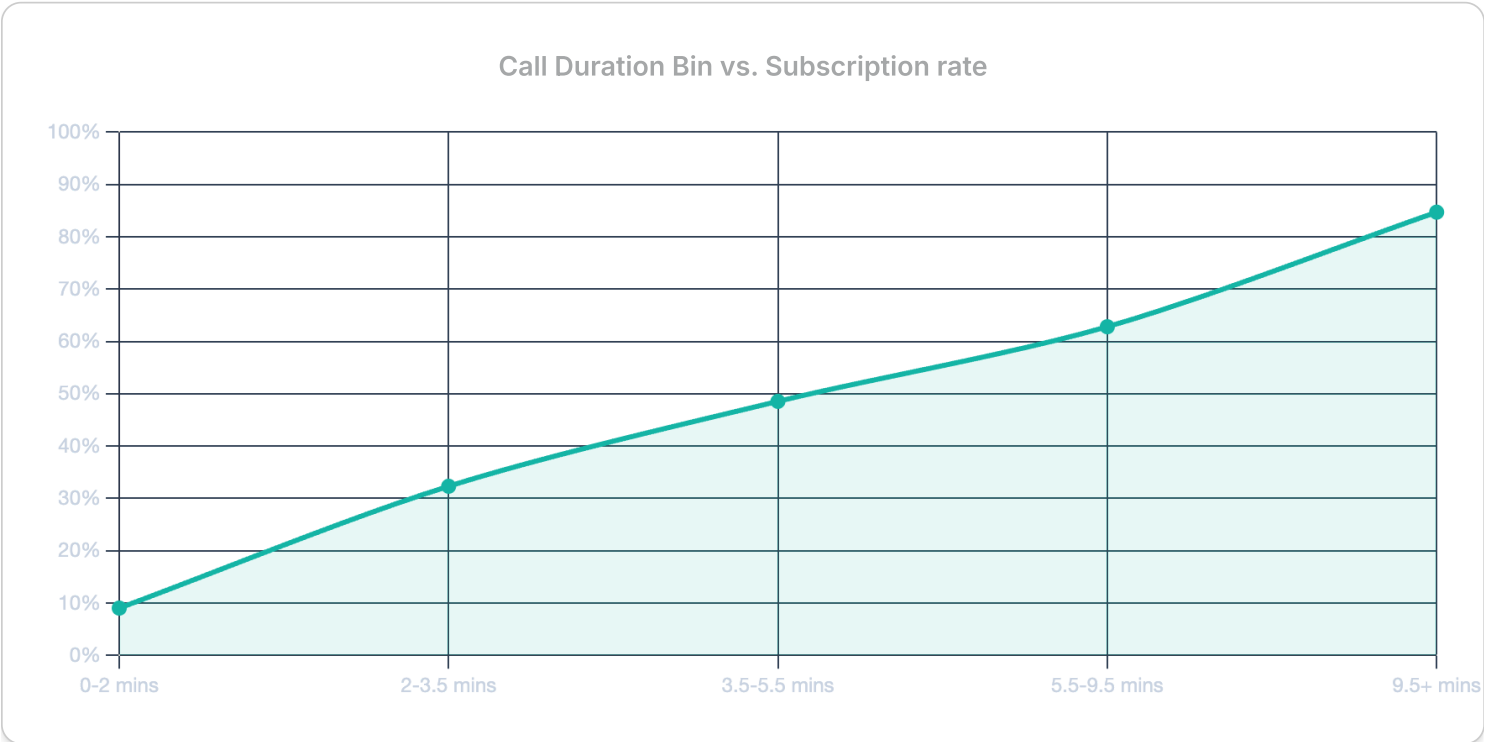


As shown above, **Call Duration** is the single most dominant predictor (36.87% relative importance), followed by **Balance** (8.88%), **Age** (8.40%), and **Month** (8.32%).

Deep Dive on Key Drivers

1. The Call Duration Effect (Golden Metric)

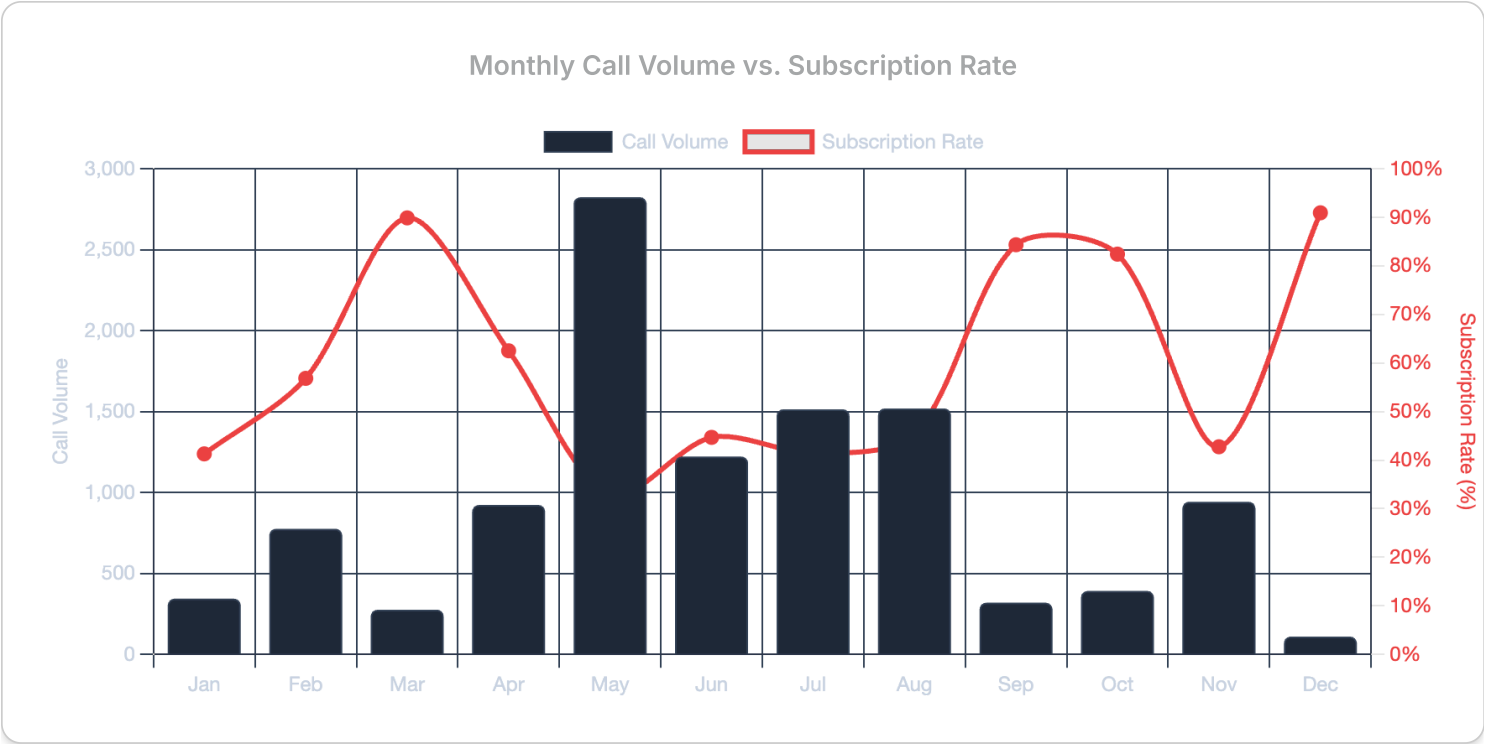
Call duration has a direct, positive correlation with subscription. Long conversations give agents time to address concerns, outline interest rate calculations, and close the sale.



- **0-2 mins:** Converts at only **9.02%** (represents immediate hang-ups or failed initial pitches).
- **3.5-5.5 mins:** Converts at **48.54%**.
- **9.5m+ mins:** Converts at a staggering **84.70%**.

2. The "May Campaign" Bottleneck

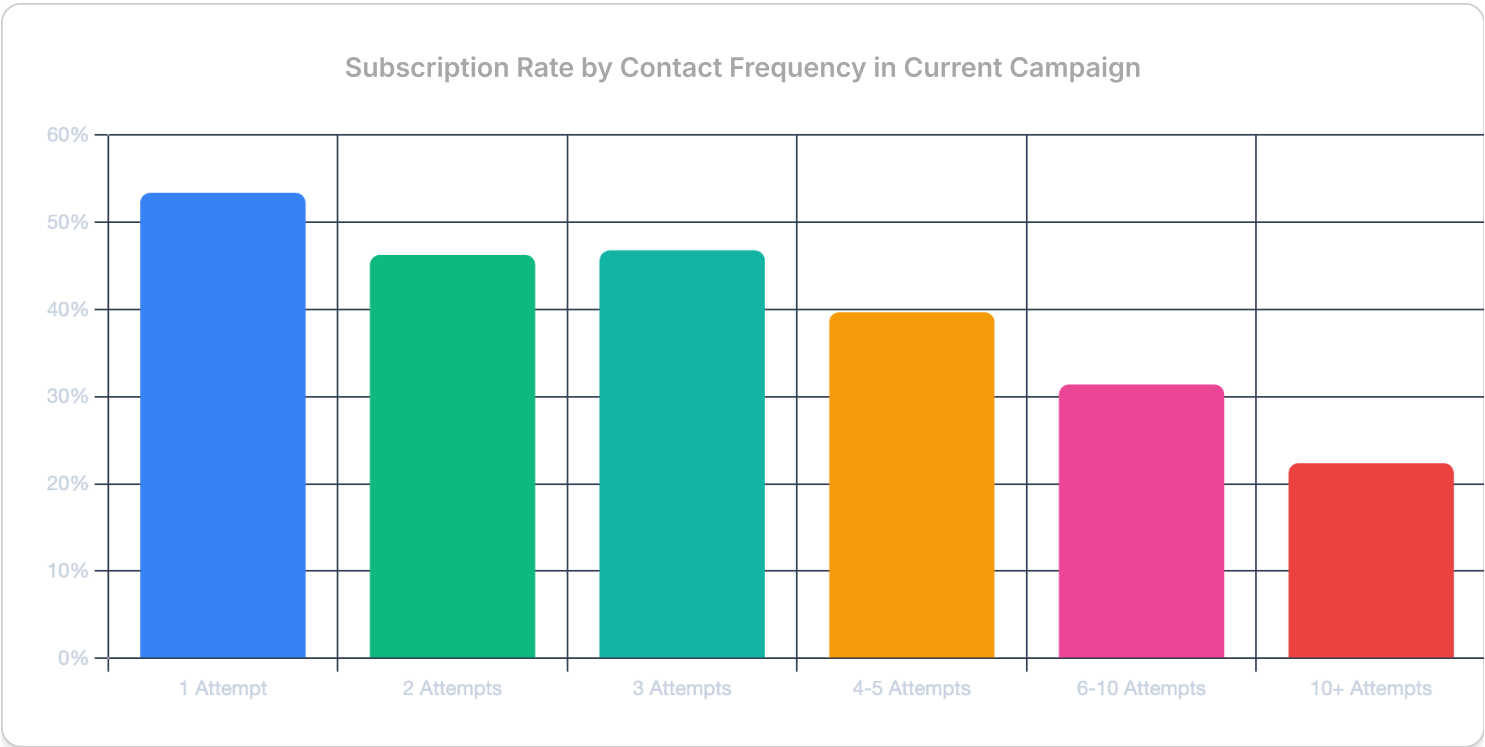
We discovered a major misalignment between marketing volume and conversion rates across different months of the year:



The bank is pouring massive resources into May (2,824 calls, 25.3% of total volume), yet May records the lowest subscription rate of the year (**32.76%**). In contrast, transitional months like March (89.86% conversion), September (84.33%), and October (82.40%) are highly underserved but convert at exceptional rates.

3. Diminishing Returns of Contact Frequency

Re-calling the same customer during a campaign yields sharp diminishing returns. Our analysis shows that after the 3rd attempt, conversion rates collapse:



While the first contact converts at **53.38%**, the rate drops to **31.40%** for 6-10 contacts, and down to **22.38%** for 10+ contacts. Calling customers more than 3 times is an operational waste and causes brand annoyance.

5. Business Insights & Segmentation

Using demographic and financial cross-tabulation, we identify target opportunities and high-risk profiles:

High Opportunity Customer Segments

- **Students & Retirees:** Students convert at **74.72%**, and retirees convert at **66.32%**. These cohorts have high interest in simple, low-risk interest-bearing accounts.
- **High Account Balance:** Customers in the top quintile (>€1,708 balance) convert at **58.00%**, showing they have the liquidity to commit capital.
- **Prior Warm Leads:** Customers with a successful prior campaign outcome (poutcome = success) convert at ****91.32%**** in the current campaign.

High Risk / Avoid Segments

- **Borrowing/Debtor Clients:** Customers with personal loans convert at only **33.15%** and housing loans convert at **36.64%**. These clients are using cash to pay debt, leaving no room for savings deposits.
- **Blue-Collar Workers:** The bank placed 1,944 calls to blue-collar workers, but they convert at a below-average rate of ****36.42%****.
- **Unknown Contact Channels:** Calls where the contact method (e.g. cellular, telephone) is unknown yield only ****22.59%**** conversion.

6. Actionable Recommendations

Based on our statistical analysis, we recommend the following five strategic initiatives:

1. Re-Allocate Monthly Campaign Budgets

Reduce outreach volume in May by 50% and shift call center capacity to March, April, September, and October. Shifting resources to these high-performing transitional months will capture clients when they are more receptive to savings offers, increasing campaign ROI.

2. Implement a Smart Call-Capping Policy

Enforce a hard cap of **3 contact attempts** per customer per campaign. If a customer is contacted three times without subscribing, they should be flagged as inactive and removed from the calling queue. This will save call center resources and prevent customer fatigue.

3. Segment-Specific Product Pushing

Create separate campaigns for target segments:

- Pitch low-risk, predictable steady-yield products to **Retirees**.
- Pitch high-yield, flexible-tenure micro-savings products to **Students**.
- Avoid cold-calling debtors with deposit offers; instead, cross-sell loan refinance products.

4. Enforce Database Contact Enrichment

Prioritize cellular contact channels (54.33% conversion) and enrich the client database to eliminate the "unknown" contact channel classification (22.59% conversion). Update customer records during regular app logins or branch visits.

5. Automate Warm Lead Priority

Automate immediate calling queues to target prior campaign successes (outcome = success) at the start of any new campaign. These clients are highly loyal and represent low-hanging fruit with a 91.32% subscription probability.

7. Conclusion

By moving away from unsegmented high-volume cold-calling (epitomized by the massive May campaigns) and transitioning to target segments (students, retirees, high-balance clients, and warm leads), the bank can achieve higher conversion rates with a fraction of the current calling volume. Implementing contact capping and database enrichment will improve campaign efficiency, reduce acquisition costs, and maximize deposit revenues.